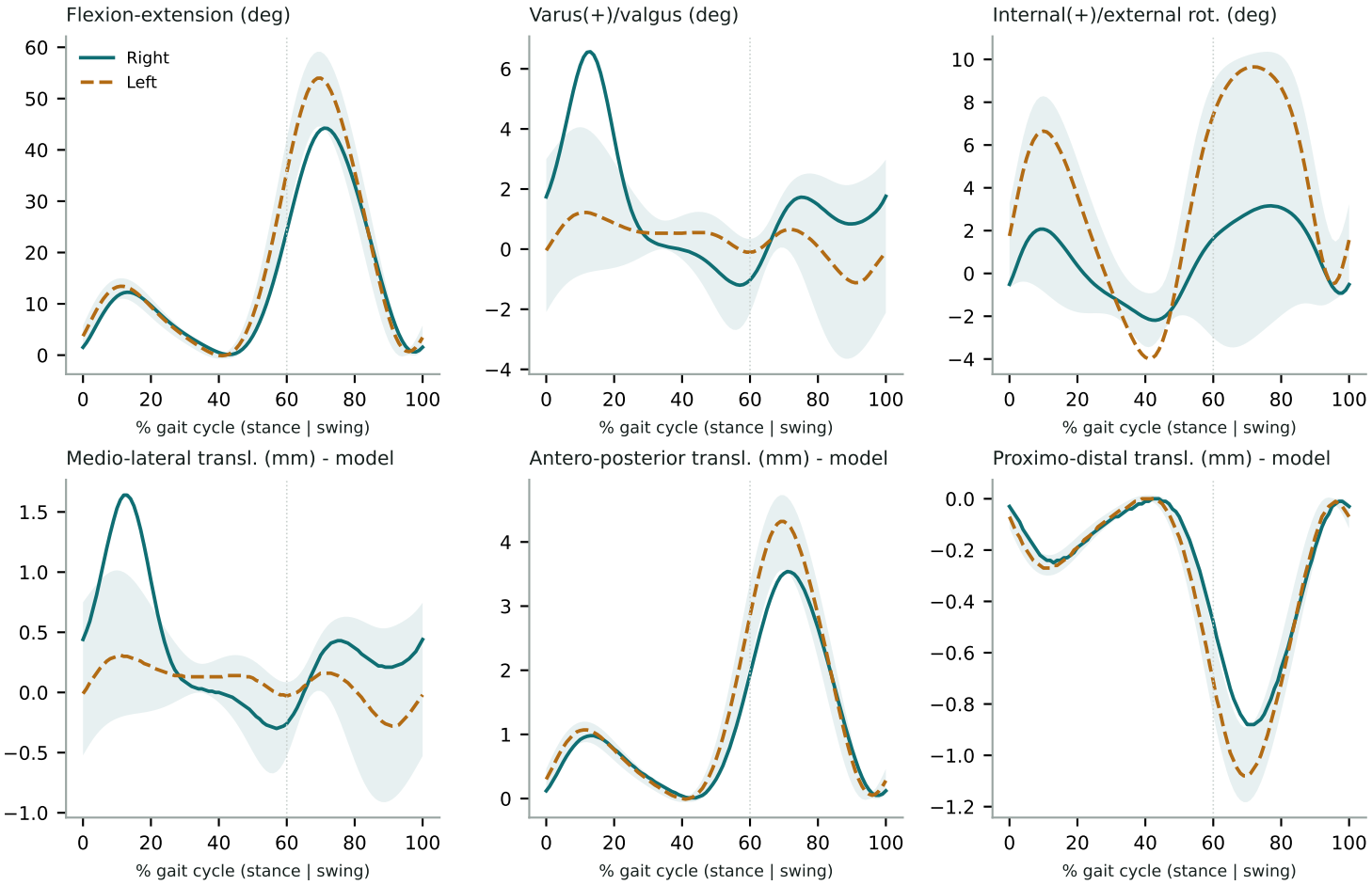
GaitLens-6DOF | Knee kinematics report

Subject: VIRTUAL-OA-100 (synthetic) Protocol: 3 x 10 m walk, self-selected speed Sensors: 4 IMU + side camera Session date: 15 Oct 2026

Right knee shows an early-stance varus thrust of 4.8 deg (healthy 1.4 +/- 1.6), consistent with medial-compartment loading. Right peak swing flexion is reduced to 44 deg (-20% vs left), a stiff-knee swing pattern. Left knee is within healthy ranges except a borderline rotation range. Stride time 1.19 s.

MEASURE	RIGHT	LEFT	HEALTHY RANGE	STATUS R / L
Peak swing flexion	44.2 deg	54.0 deg	52.1 +/- 7.4	WATCH NORMAL
Loading-response flexion	12.3 deg	13.4 deg	13.1 +/- 2.0	NORMAL NORMAL
Varus thrust (early stance)	4.8 deg	1.3 deg	1.4 +/- 1.6	FLAG NORMAL
Rotation range (IE)	5.3 deg	13.7 deg	6.8 +/- 6.4	NORMAL WATCH



Shaded: healthy reference band (mean +/- 2 SD). Dotted line: typical toe-off.

Method: four skin-mounted IMUs (thigh and shank, both legs) with functional two-pass axis calibration, heading-drift correction from standing pauses and stride stationarity, flexion-varus crosstalk minimisation and 6 Hz zero-phase filtering. Rotations follow the Grood-Suntay joint coordinate system relative to quiet standing. Translations are estimated from a tibiofemoral coupling model driven by the measured rotations and are not independent measurements; do not use them to assess ligament laxity. Expected accuracy (virtual validation, 64 knees): flexion 1.8 deg, varus 1.8 deg, rotation 2.6 deg RMSE. Peak flexion reads ~4 deg low due to soft-tissue artefact; compare against this device's own reference band only. Minimal detectable change (95%): varus thrust 0.7 deg, peak swing flexion 2.5 deg.

Research prototype output generated from simulated data. Not a diagnostic device.